

Unless stated otherwise, all rings are assumed to have the normal addition and multiplication operations.

Part 1

1. [3 pts each] Provide the definition for each of the following:
 - a. Integral domain
 - b. Onto function
 - c. Ring isomorphism
 - d. Ideal

2. [3 pts each] Provide an example for each of the following (do not use an example from problems 1-15):
 - a. A ring without identity
 - b. A non-commutative ring
 - c. A field
 - d. A function which is not 1-1
 - e. An onto function
 - f. Two rings which are isomorphic
 - g. Two rings which are not isomorphic
 - h. An ideal

Part 2

[9 pts each] Choose the indicated number of problems from each section.

Rings: choose any three.

1. Complete the following:
 - a. Give an example of a subset of the integers which is not a ring (under the usual addition and multiplication).
 - b. Give an example of a subset of the integers which is a ring.
2. Prove that $\mathbb{Q}$ is subring of $\mathbb{R}$.
3. Complete the following:
 - a. Prove that the set of odd integers is not a subring of $\mathbb{Z}$.
 - b. Prove that in general the union of any two rings is not a ring.
4. Prove that the intersection of any two rings is also a ring.
5. Let R be a ring and $a, b \in R$. Prove that $(-a)(-b) = ab$. Hint: $(a + (-a))(-b) = 0$.

Integral Domains and Fields: choose any two.

6. Prove that $\mathbb{Z}_n$ is an integral domain if n is not prime.
7. Prove that $\mathbb{Z}_3$ is a field.
8. Let R be an integral domain and suppose that $ax - bx = 0$. Prove that if $a \neq b$ then $x = 0$.
9. Let R be an integral domain and let $a, b, c \in R$ where $c \neq 0$. If $ac = bc$ then $a = b$.
10. Let R be a field. Prove that R is an integral domain.

Homomorphisms, isomorphisms, and ideals: choose any three.

11. Prove that $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = e^x$ is one-to-one but is not onto.
12. Let $f: R \rightarrow S$ and $g: S \rightarrow T$ for rings R, S , and T . Prove that $f \circ g: R \rightarrow T$ is a homomorphism.
13. Let $f: \mathbb{Z}_4 \rightarrow \mathbb{Z}_4$.
 - a. If $f(x) = 2x$ prove that f is not a homomorphism.
 - b. If $f(x) = 3x$ prove that f is a homomorphism.
14. Prove that $f: \sqrt{2}\mathbb{Z} \rightarrow \sqrt{2}\mathbb{Z}$ where $f(a+b\sqrt{2}) = a-b\sqrt{2}$ is an isomorphism ($\sqrt{2}\mathbb{Z} = \{a+b\sqrt{2}: a, b \in \mathbb{Z}\}$).
15. Show that any ring isomorphism $f: \mathbb{Z} \rightarrow \mathbb{Z}$ must be the identity map.
16. Let A and B be ideals of R . Prove that $A \cap B$ is an ideal of R .
17. Prove that $\mathbb{R} \times \mathbb{R} = \{(a, b): a, b \in \mathbb{R}\}$ is isomorphic to $\mathbb{C} = \{a + b\sqrt{-1}: a, b \in \mathbb{R}\}$ if multiplication in $\mathbb{R} \times \mathbb{R}$ is defined as $(a, b)(c, d) = (ac - bd, ad + bc)$ and addition is defined as $(a, b) + (c, d) = (a + c, b + d)$.